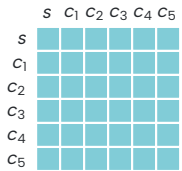
Shared: state s · $K = 5$ candidates $\{c_i\}$ sampled by the frozen SFT actor · one pick executed

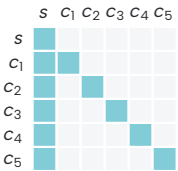
a) ADM head on frozen actor features

Joint

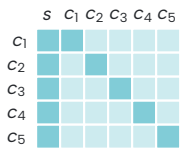
Independent

Transformer (2 layers)





MLP (depth 2)



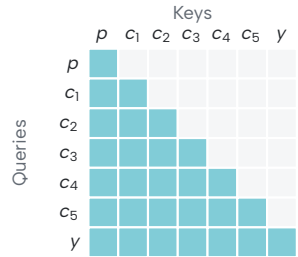
MLP input $[c_i,\ s,\ c_i - \overline{c}_{-i}]$

MLP input $[c_i,\ s]$

logit z_i per candidate · softmax over K · argmax
reads actor hidden states (layers 18/27/36) · 22.6 M trained

b) ADM-4B decoder

LoRA on a separate OpenWebRL-4B-SFT copy



p: task, history, URL, screenshot
c_i: numbered candidate (action + reasoning)
causal: c_i reads only earlier candidates

y generates {"selection": i} (JSON schema)
separate 4B pass over prompt + candidates

Allowed · Mean of the others only · Masked